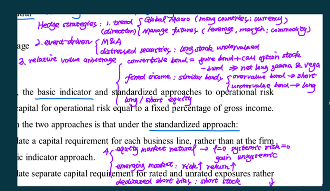
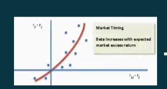
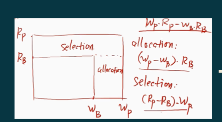
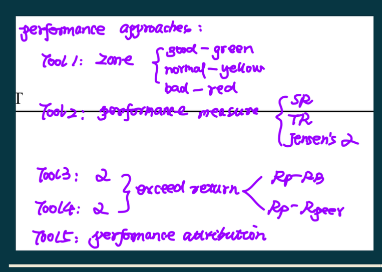
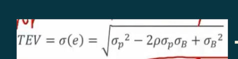
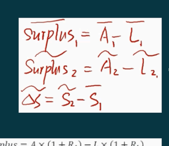
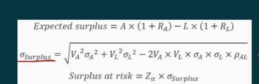
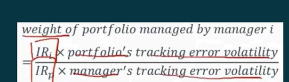
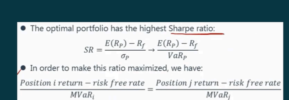
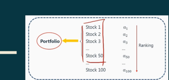
Risk & Investment Management

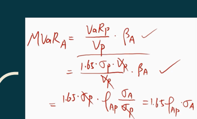
Hedge Fund
- Introduction
  - Little transparency
  - Award
    - Bonus
    - Fixed management fee
  - Overstate return, SR, autocorrelation
  - Understate risk, beta, volatility
  - Bias
    - Survivorship bias
    - Self-selection bias
    - Backfill bias
- Evolution of the hedge fund industry
- Hedge fund strategies
  - Trend/Direction
    - Global macro
      - Many countries
      - Currency
    - Managed futures
      - Leverage & Margin
      - Commodity
  - Event Driven
    - Merger arbitrage — Short bidder stock Long acquired firm stock
    - Distressed securities — Long undervalue stock
  - Relative Value Arbitrage
    - Convertible arbitrage
      - CO = Bond + Call on Option
      - Short Bond and Equity
      - Net long gamma & vega
    - Fixed income arbitrage (Similar bonds)
      - Short overvalue stock
      - Long undervalue stock
    - Long/Short Equity
  - Other Strategies
    - Equity market neutral — Beta = 0
    - Emerging market
    - Dedicated short bias
  - Funds of hedge fund — High transaction
  - 
- Risk management of hedge fund
- Due diligence
  - Factors
    - Risk control
    - Business model
    - Investment process

Performance Measure and Evaluation
- Performance measure
- Performance evaluation
  - Statistical significance of alpha
    - $N > \left(\frac{1.96}{IR}\right)^2$
  - Market timing ability
    - 
  - Performance attribution
    - Allocation
    - Selection
    - 
- 

Portfolio Risk Management
- Absolutely & Relative risk
  - Relative risk -- relative to benchmark
    - $TE = R_P - R_B$
    - 
- Measure different risk
  - Policy mix risk
  - Active management risk
  - $R_{asset} = R_{policy\ mix} + R_{active\ mgt}$
- Funding risk Surplus
  - 
  - 
- Sponsor risk
  - Cash flow
  - Economic risk-investment risk
- Risk budgeting
  - Determine total VaR
  - Choose optimal allocation of asset
  - 
  - 
- Liquidity consideration
  - $LD = \frac{Q}{(0.10 \times V)}$

Factor Investing
- Explanation
  - Rational explanation
  - Behavior explanation
- Factors
  - Macro factors
    - Economic growth
    - Inflation
    - Productivity risk
    - Demographic risk
    - Political risk(sovereign risk)
    - Volatility
      - CAPM
      - Volatility increase return increase price decrease
      - Mitigating volatility risk
        - Volatility swaps
        - Out-of-the-money put
          - Like volatility buy option -- sell volatility protection
          - Low price
  - Investment factors
    - Static factors
    - Dynamic factors
      - Fama and French
        - Market
        - SMB-- long small cap & short big cap — This is not efficient all the time.
        - HML-- Short growth stock & long value stock
          - Value stock has high and asymmetric adjustment cost
          - Negative feedback is a method to stabilise the market
          - Always being efficient
      - Momentum — Winner - Loser
        - Positive feedback
        - **Destabilise the market** （**Amplify trends / Fuel momentum**）
- Low-Risk Anomaly
  - Definition--Low volatility & Low risk -- high return
  - Lagged volatility & future return
  - Lagged beta & future return
- Alpha
  - Jensen's alpha = Rp-Rcapm
  - Alpha = Rp-Rbenchmark
- Fundamental law of active management
  - $IR \approx IC \times \sqrt{BR}$
  - $IR = \frac{R_P - R_b}{\sigma(R_P - R_b)}$

Portfolio Construction
- Inputs
  - Refining alpha — Benchmark -- alpha = 0
  - Transaction — Rebalancing -- rebalance cost
    - *more frequently, more stable but more trading cost*
  - Proper alpha coverage — Expanding benchmark
- Techniques
  - Screens
    - 
    - Cons — Simple
    - Pros — Excludes some categories
  - Stratification — Solve the bias of categories
  - Linear programming
    - industry, size, volatility, beta
    - The number of the stocks is not sufficient.
  - Quadratic programming
    - Alpha, risk, transaction cost
    - More inputs and more noise

Portfolio Risk Measures
- Diversified VaR — Portfolio VaR
- Underversified VaR — VaR1, VaR2
- Marginal VaR
  - 
- Incremental VaR
  - New position
  - $\text{Incremental VaR} \approx VaR_{P+a} - VaR_P$
- Component VaR
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